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*** START OF THE PROJECT GUTENBERG EBOOK THE
IMPOSSIBLE INVENTION ***

The Impossible Invention

By Robert Moore Williams

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I had to admire this little guy's courage. Fradin, his name was—James Arthur Fradin, with a string of letters after it that even the alphabet agencies down at Washington could not have unscrambled. The letters represented honorary degrees conferred on him by half a dozen different colleges, and they should have entitled him to be heard with respectful consideration, but they weren't. The assembled scientists of the Institute of Radio Engineers were giving him merry hell.

"What you are saying, Fradin," one of the scientists interrupted hotly, "is gross nonsense."

"It is absolutely impossible," another shouted.

"Faker!" somebody yelled, and a dozen voices took it up until the room echoed with the sound.

I sat back and grinned to myself. If this meeting ended in a free-for-all fight, which was what looked like was due to happen, I would be able to make a swell human interest humorous yarn out of it. My editor went for human interest stuff, which was largely why he had sent me down to cover this meeting. He knew I wasn't likely to develop any front page news here, scientific meetings being what they are. But there might be a human interest angle that would be good for a laugh. And the way these solemn scientists were calling Fradin a liar, it looked like the laugh was coming.

There was one man who wasn't doing any name calling, I noticed, a tall, cadaverous-looking individual sitting two seats down from me. He had listened very carefully, almost eagerly, I thought, to everything the speaker had said. Glancing at him, I got the impression that I should know him, but at the moment I couldn't place him. Tall, bony face, thin, hawk nose—yes, it seemed I should know him.

Fradin had stopped speaking when the storm of abuse broke over him. He stood there on the platform, a little, white haired guy with a gentle face.

"If you numbskulls will only be quiet for a moment," he said, when the noise had subsided for an instant, "I will offer incontrovertible proof to support my statement that radio waves are transmitted through what I must,

for lack of a better term to describe the undescrivable, call the fourth dimension."

What I mean, the roof must have been nailed down tight, or the explosion that followed would certainly have lifted it off the building. You never did see so many excited scientists in one group. Normally a scientist is supposed to be cool, aloof, and impersonal. But this group was anything else! They went right straight up in the air. I couldn't tell whether they were angrier because he had called them a bunch of numbskulls or because he had said that radio waves were transmitted through the fourth dimension.

One of them leaped to his feet. Ramsen, I think his name was. He was a big shot in the field, almost as big as De Forest and Marconi.

"Fradin," he yelled, "that is the most preposterous statement I ever heard from the lips of any man in his right senses. It raises the immediate question of whether or not *you* are in your right senses."

There was a buzz of approval following his statement. Fradin waited for it to die down.

"Mr. Ramsen," he said, "you have chosen to challenge my theory. Perhaps you can tell me what medium *does* carry the electro-magnetic radiations that we call radio waves?"

"Certainly," Ramsen answered. "Any schoolboy knows that."

"We are not here concerned with the knowledge of schoolboys," Fradin gently replied. "Sound is carried in air and water and by many solid substances. But we know that radio waves do not travel in air, because they will pass through a perfect vacuum. In what medium do they travel, Mr. Ramsen?"

Right here was where I began to pay close attention. Something about Fradin's manner, his calmness, his certainty, gave me the impression that he knew pretty much what he was talking about.

"Radio waves, Mr. Fradin," Ramsen answered, in the manner of a scoutmaster revealing the facts of life to an errant Boy Scout, "travel in the

ether."

He was right. I was not assigned to cover this meeting by chance but because I happen to have a pretty good groundwork in science. Radio waves, all scientists admit, are propagated through the ether.

"And what," Fradin countered, "is the ether?"

"Why—" Ramsen answered. "It's—" He started to flounder. A sudden silence fell in the room. Ramsen's face started to get red. "The ether," he finished, "is—why it's the ether, that's what it is."

"What you are refusing to admit," Fradin crowed, "is that 'ether' is a meaningless word invented by numbskulls such as are gathered here to describe something about which they know absolutely nothing. The ether is a word, nothing more. It does not exist. The Michelson-Morley experiments conclusively proved that, if it existed, its nature was such that it could not be detected by any physical experiment whatsoever. In other words, that it exists only as a handy tool by which scientists who ought to know better can conceal their own ignorance. Gentlemen," he said, turning from the red-faced Ramsen to the perturbed audience, "I can not only conclusively prove that radio waves are transmitted through the fourth dimension—but I can also prove that power, actual power, can also be transmitted through the same medium!"

He stopped suddenly, biting his lips as if he had said more than he had intended to. But I think only one man in the audience had caught the full implication of Fradin's words. The rest of them were too busy defending themselves against the accusation of being numbskulls to notice the one really important thing he had said.

How that audience did boil! And they boiled because every man jack of them, in his heart of hearts, knew that Fradin was right. I knew it the minute he said it. And they knew it too. When he said that "ether" was only a word used by fools to conceal their own ignorance, he had hit the nail exactly on the head.

For that is precisely what it is. Nobody has ever seen the ether, felt it, smelled it, heard it, touched it. Scientists of the past century, needing a mechanical device to account for the observed propagation of electromagnetic radiations such as light and the then little known radio waves, had

invented the ether to carry those radiations, invented it out of whole cloth. Fradin's hearers knew he was right. Taken individually, when they were calm, they would have admitted it. But they were in a group and he was calling them fools right out in public. Mass hysteria got them. They boiled over and very promptly demanded that he prove his statements.

He refused to do it. Absolutely refused.

"We demand that you produce your proof," Ramsen howled. "You have called us fools and said you could prove it. We demand that you *do* prove it."

"I—" Fradin began. He wet his lips. His face had whitened. It wasn't a gentle, kindly face any more. It was the face of a badly scared man.

Fradin was scared. But he wasn't scared of those engineers who were shouting at him. He was scared of something else.

"Speak up," Ramsen roared. "Produce your proof!"

"I can show you mathematical proof," Fradin offered.

How they howled at that, all except the tall, thin, hawk-nosed individual sitting two seats down from me. He took no part in the demonstration. Instead he got up and very quietly went out of the room.

As he walked out, I again got the impression that I ought to know him. But I still couldn't place him. A reporter sees too many people to remember all of them.

"Mathematical proof, unless supported by incontrovertible experimental evidence, is not sufficient," Ramsen thundered. "We demand that you produce experimental proof."

By experimental proof, he meant an actual instrument of some kind to demonstrate Fradin's claims—some gadget that they could see and feel and examine, something they could take apart and put back together again, something that they could watch in operation. Ramsen was quite right in

making such a demand, for without experimental evidence to back it up, mathematical theory is more often than not just so much hot air.

"Your demand is just," Fradin faltered. "I fear in the heat of argument I made statements I do not care to support. I do not choose to produce the experimental evidence that I have."

He didn't say another word. Instead, he turned and walked off the platform, going out through a door at the back. Nor did he enter the lecture hall where the meeting was being held. He walked out of the room.

And he didn't come back.

Why had he refused to produce the evidence that he had? What had scared him?

Questions were buzzing like gadflies in my mind. There was one particularly persistent gadfly. Fradin had said something of which I had almost caught the significance. Almost, but not quite. The significant thing he had said kept buzzing in the back of my mind, but I couldn't quite put a mental finger on it...

Then I remembered it.

I went out of that room at a dead run. I went up over the speaker's platform and through the door Fradin had taken. How I did want to talk to that tortured man!

What he had said, letting it slip accidentally, added up to one of the biggest stories that ever splashed across the front page of a newspaper. I had come down here looking for a human interest yarn. Instead I had run straight into a story that could easily set the world on fire, *if* I could find Fradin and make him talk.

I didn't doubt that I would find him. He couldn't have gotten far away. He hadn't had time. Not five minutes had passed after he had walked off the platform until I was following him.

The door opened into a long hall, and in that hall I found Fradin. He was down at the far end, getting into an elevator. A tall, thin individual was with him.

I sprinted down the hall to try to catch them before the elevator doors closed. The operator saw me coming and started to wait for me, but something changed his mind for him. The two men were already in the cage; I couldn't be positive of it, but I thought the tall man said something to the operator. Just as I got there, the operator slammed the door in my face. The cage started down.

"All right, smart guy," I yelled at the operator. "I'll give you a smack in the snoot for this."

Probably I could have gotten down faster by waiting for another elevator, but there was a stairway and I used that. I was in a hurry.

I got to the first floor just in time to see Fradin and his companion walk out of the front door.

"Hey!" I yelled. "Wait—"

I started to say, "Wait for me," but the words were choked off in my throat. I recognized Fradin's companion. The hawk-faced man who had sat two seats down from me and who had slipped unobtrusively out of the meeting. He had gone around to the back of the hall and joined Fradin.

But the fact that he was with Fradin wasn't the thing that had choked off my call. It was the way the two men were walking. Fradin was a little ahead, and he wasn't looking to the right or the left. Just walking. There was a stiffness about him that made me think of a mechanical toy that has been wound up and is taking a walk for itself.

Hawk-face was following right behind him. Hawk-face had his hand in his pocket. I didn't need to look twice to know that he had a gun in that pocket too, pointing straight at the little inventor's back.

It wasn't a stick-up. It was a kidnap job. Hawk-face had also heard the really important thing that Fradin had said in his speech. I had missed it for a few minutes, but he hadn't missed it. He had instantly realized how damned important it was. He had walked out of the meeting, gone around to the back, waited for the scientist.

I should have called the police. But just then something happened that upset me so badly I completely forgot all about police.

I recognized Hawk-face, and cold chills began to run up and down my spine.

His name—or at least the name attached to the picture I had once seen in the hands of an F.B.I. man—was Marvak. The name didn't mean anything. He had others. A name to use in Asia, one for Russia, one for Europe, all different.

Marvak was one of the names he used in America, but the F.B.I. suspected he had others. They would cheerfully have hung him by any of his names, if they could have caught him, but he didn't catch easily. Compared to him, an eel was a rank amateur in slipperiness, and a rattlesnake was man's best friend.

Right out in front of the building, on a crowded city street, he forced Fradin into a cab. I was so close I could see the haunted, terrified expression on the little scientist's face. But I didn't think then and I know now that he wasn't afraid of the gun. He was afraid of something even more horrible than Marvak.

The cab pulled away.

An eternity seemed to pass before I could collect my faculties and grab the next cab in line.

"Follow that cab," I hissed at the hacker. "There's a ten-spot in it for you if you don't lose it."

He didn't lose it. We followed Fradin and Marvak down to an old, abandoned factory building on the outskirts of the city. They were getting out of their cab when we drove up. Marvak, with his right hand still in his coat pocket, was paying off the driver.

"Drive on past," I told my driver.

He went on past and around the block and I got out, but the driver had to remind me about the ten-spot I had promised him. I paid off like a slot machine.

That shows how excited I was. I had stopped thinking about the story I might get. The story was secondary now. What really mattered was that Fradin had to be rescued, and fast. The thing he had let slip was too big to fall into Marvak's hands under any circumstances. And I had to be the one who rescued him. There wasn't time to go for the police. Marvak would work fast.

Marvak did work fast.

I tried the front door first because that was the obvious thing to do. If I got caught coming in the front door I could say I had the wrong address and back out. But if I got caught at the back door, no amount of explanation would do me any good.

The front door was unlocked. It opened into what had once been an office. A flight of stairs led up to the second floor.

I listened—there wasn't a sound. "They went upstairs," I thought.

"Hold it, bud," a voice said behind me.

The voice had a chilled steel ring that sent my heart right down into my shoes. There was no mercy in it, no compassion, no pity. It was double-edged with the threat of death. It jerked my head around.

Marvak was standing there in the office. He had the gun out of his pocket now, pointing it straight at me.

There was a closet in the office. Marvak had simply waited in the closet until I entered. When I started up the stairs, he had stepped out behind me.

"I—I must have got the wrong ad—address," I faltered.

His eyes, gray chilled steel, they were, drilled into me.

"No doubt," he said—but very doubtfully. "You're the reporter I noticed at the meeting, aren't you? What are you doing here?"

How big a lie could I tell and still be safe? How close to the truth could I come and not get one of those slugs between my eyes?

"I came out to interview Mr. Fradin," I blurted out.

He seemed to let it go, but back of those cold gray eyes I could see his mind working as he decided what to do with me. Then I saw him reach a

decision.

First, he searched me. I didn't have a gun, which seemed to surprise him.

"You can come along," he said. "If Fradin can demonstrate to my satisfaction the discovery he claimed to have made, it will make the headlines, *if* you get a chance to write it."

With that, he dug the little inventor out of the closet, and with the gun out, prodded both of us upstairs. There were only two floors to this building and the entire second floor was Fradin's laboratory.

It was crammed to the ceiling with the weirdest collection of electrical equipment I have ever seen. Generators, dynamos, electric motors. There was enough radio equipment to set up a modest broadcasting station. And in one corner was something big enough to be a cyclotron. Fradin had just about everything in his laboratory.

"Now," said Marvak to the little inventor, "you will please prove the truth of your assertion that power can be transmitted by radio."

That was the thing that Fradin had said. Power by radio! It doesn't sound like much, but let me tell you, it's plenty big. With it, science could come darned close to remaking the face of the globe.

How?

This is the power age. Practically all of our industrial achievements—and through them we have achieved what passes for civilization—have come about through cheap power. Coal, the steam engine, the dynamo, water power. Maybe, not so long in the future, we'll have atomic power, but we don't have it yet. All we have now are coal and water. And possibly 90% of the water power in this country and probably 95% of the water power on earth are going to waste, simply because the waterfalls are usually in mountains and the places where the current is to be used are in cities hundreds and even thousands of miles away. *Transmission losses over high lines are so great that electrical energy cannot be efficiently transmitted very far.* So the water power goes to waste.

But here we have radio transmitted power. No high lines, hence no high line losses. Of course there would be other losses, but if Fradin said power could be transmitted by radio, he would know how to cure the losses. Radio transmitted power would make electricity so damned cheap that every home in the country could have it.

And this is only part of the picture that Fradin's invention brought into being. Supposing power could be transmitted by radio. Suppose automobiles could pick it up and use it. Then the extremely expensive internal combustion engine that goes into every car could be replaced by cheap motors. The price of cars could be cut in half. Everybody could have one. And operation costs would be next to nothing.

Ocean liners? No more bulky, costly steam engines. Boats could take their power out of the air.

And airplanes. There was the most important item of all. No gasoline engines in planes, no engine failures, no crashes because the motor conked out. Air flight spanning the globe.

That's what radio transmitted power ought to mean, that's what it would mean—until Marvak entered the picture. When he appeared on the scene, power by radio, instead of being a blessing, would become one of the worst disasters that ever happened to humanity.

Marvak was a spy. Not a common, garden variety of spy, not a fifth columnist, not a saboteur, but a sort of super-spy who sold his services to the highest bidder. If you wanted a war started, he could make all the arrangements to provide for an "incident." If you wanted to take over a minor nation, he could pave the way for you; if your enemy had a new and secret weapon, he could get the plans. Anything, just so he was well paid for it.

If Fradin could really transmit power by radio, and if Marvak got the plans, the waterfalls would not be harnessed, there wouldn't be cheap automobiles, and handy power for ocean liners. There would be power—unfailing power—for one thing: planes! Bombing planes, fighting planes!

If you think several nations on this globe would not jump at the chance to acquire such an invention, you have another think coming. And the price they would be willing to pay for it, would be big enough to interest even

Marvak. It would be worth—well, what is the worth of the British Empire, China, and the United States?

Fradin's invention had exactly the same value as those three nations lumped together, if Marvak succeeded in peddling it in Europe. Bombers over New York, bombers over Chicago. There would no longer be any safety in three thousand miles of water. Bombers over London. New bombers that would be almost invincible.

Sweat was running down over my face, down over my body, down over my soul. If Marvak got Fradin's invention, Johnny Holmes—that's me—go hunt for an air-raid shelter, because you're sure going to need it.

"I was mistaken," Fradin faltered, his voice a whisper. "I was—boasting. I cannot transmit power by radio."

"You're a liar!" Marvak snapped.

"I'm not a liar," Fradin whispered.

"Either or else," Marvak said, bringing up his gun until it pointed right at the little inventor's forehead.

Fradin had something that a man could call courage. He looked that gun in the eye. His face went a shade whiter, but his eyes did not drop.

"I'm afraid it will have to be else," he said. As he spoke the words, he seemed to stiffen himself until he stood very straight. He looked like a soldier standing at attention. "But if you shoot me you may find it difficult to operate my invention."

Marvak's finger tightened around the trigger. His face was cold with rage, his grey, killer eyes looked like icicles.

"Don't shoot him, you fool!" I hissed. "Then you'll never find out what you want to know."

I was stalling for time, stalling for anything, stalling for a chance to jump that gun. I was standing beside Fradin, but the gun covered both of us.

"Shut up!" Marvak snarled at me.

The gun went off.

He had shot Fradin. It was cold blooded murder. But as he had shot the little scientist, he had taken his eyes off me. I started to jump. The gun instantly swung to cover me. I saw Marvak's face, with no mercy in it. The gun froze me motionless.

Fradin didn't fall. There was a look of surprise on his face, but he didn't fall. Then I saw what had happened. Marvak had shot him in the shoulder instead of through his head.

"That's just a sample," Marvak said, "to show you that I mean business. You're not badly hurt, but the next one will go through your knee-cap. I understand that a bullet through the knee is very painful. Now are you going to tell me what I want to know or are you going to need further persuasion?"

Blood was running down Fradin's coat. He was clutching his shoulder with the other hand, trying to stop the flow of blood. His face was very white. And now there was fear on it, fear that had not been there when he first faced Marvak's gun. I got the fleeting impression that it was not fear of the spy nor of the weapon, but of something else. I also got the impression that it was a terrible fear, a soul-consuming fear, a bleaching, whitening, shuddering fear, a fear greater even than the fear of death....

"All right," the little inventor whispered. "You win. I'll show you what you want."

"That's better," Marvak said, in a satisfied tone. "I don't mind saying that if I make a cleaning on this, I'm quite willing to cut you and the reporter in on it."

He was lying. The only way he would cut us in would be to cut our throats. Both Fradin and I knew it.

"I'm afraid," the inventor said, "that your shot has injured my arm so badly that I will have to ask you to help me."

"Okay," Marvak said. "But remember I have an excellent knowledge of electrical apparatus, so don't try any tricks, like electrocuting me by accident."

"There won't be any trickery involved here," the little inventor whispered through bloodless lips.

I watched. There were two bulky instruments, one of them a transmitter, the other a receiver. The current flow was seemingly directional. It was sent out from the transmitter and caught by the receiver. There was a meter on the transmitter to show how much current was being transmitted and another on the receiver to show how much was getting through.

There was a red line on the dial of the meter at the transmitter.

The purpose of the set-up was obviously to demonstrate that current could be transmitted by radio.

Marvak made a complete examination of the apparatus. He knew what he was doing, all right. You could tell from the way he went over the instruments that he knew his stuff.

"I'm not interested in transmitting just a little power," Marvak said. "If this thing is to be useful, it must be able to send lots of kilowatts through the ether."

"I think," Fradin answered wearily, "it will handle all the power you choose to put into it."

That was the thing Marvak had to know, that the power transmitted would be adequate to keep a plane in the air. If only a little power was transmitted, the invention, from a practical viewpoint, was useless. No dictator would give a cent for it.

Marvak handled the transmitter, Fradin tried to operate the receiver and to stanch the flow of blood from his shoulder at the same time.

Marvak turned a switch, and the power transmitter began to throb under the load. Marvak consulted the meter on the transmitter, then ran across the room to the receiver and examined the meter there.

"You've really got it!" he exulted. "There's enough power flowing through the receiver to keep a plane in the air."

I was sick, sicker than I had ever been. Fradin's invention worked. And when it worked, it spelled our doom. We would be killed, because it worked. How many millions of others would also die, I could not begin to guess.

"One plane is not enough," Marvak said. "It has to be strong enough to supply current to a fleet of planes."

He started triumphantly back to the transmitter.

"I—" Fradin faltered. He started to say something but changed his mind abruptly.

Marvak kicked over the handle of the rheostat that fed current into the transmitter. The transformer groaned. I could see the hand of the meter on the transmitter. It was moving forward as more and more current flowed into that mysterious medium that transmits radio waves.

The needle on the dial touched the red mark.

Then—it happened.

If I live to be a thousand years old I'll never be able to describe adequately what I saw happen, what I heard happen, what I felt happen. It had never happened before.

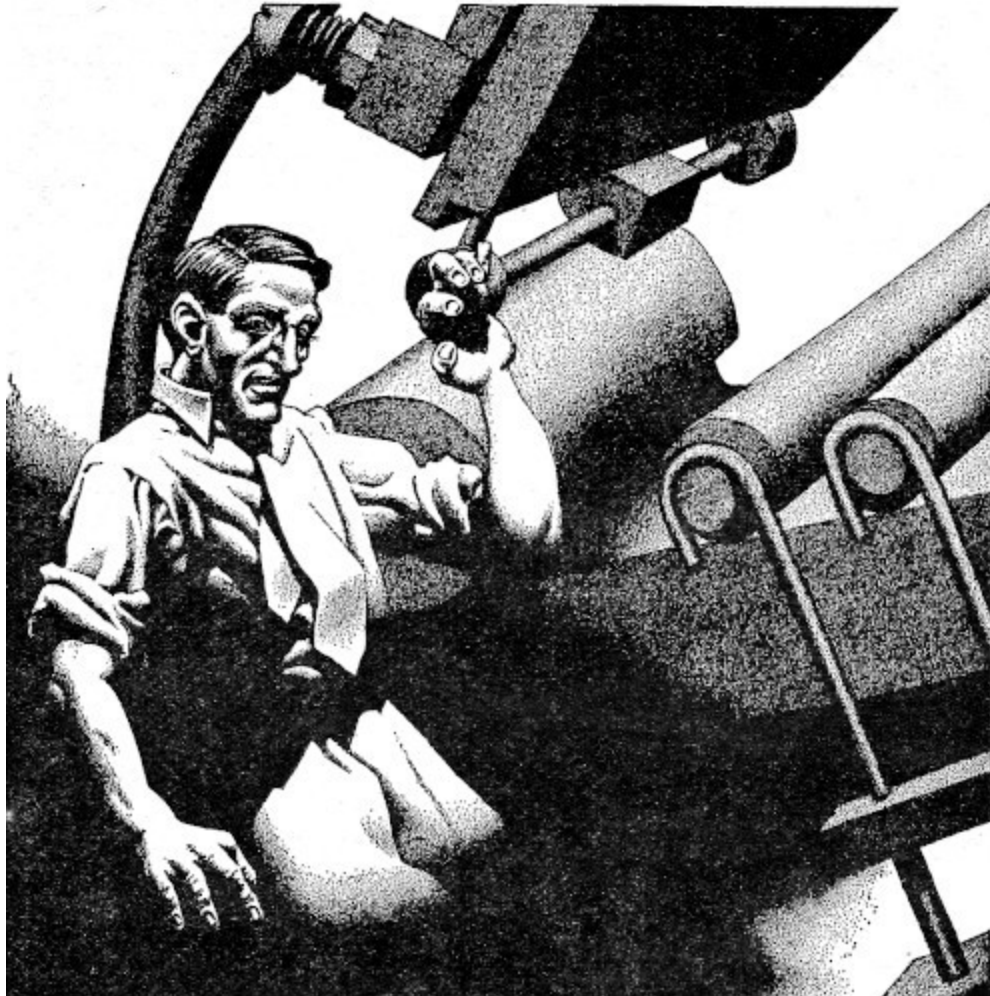
Something that I can only describe as a lightning flash ran through the room. It was a sharp, tearing crash, similar to the sound you hear when a bolt of lightning hits near you. There was a flash of brilliant light. Thunder seemed to smash my ear drums.

Fradin leaped—but not at Marvak. He leaped at me. The next instant he was grabbing me, shoving me across the room. And all the tortures of hell were breaking loose around that generator.

There was a blasting, howling roar of wind. It was the coldest wind ever. It was, I suspect, the cold of absolute zero that struck through that laboratory.

Out of nowhere, around that transmitter, a hole seemed to appear. It seemed to be torn in space. It was black, with a curiously liquid kind of blackness.

It appeared around the transmitter, and Marvak was at the transmitter. The spy seemed to freeze. A look of amazed fright appeared on his face.



Then he seemed to fall. The transmitter seemed to fall with him. Marvak tried to leap, but the footing seemed to fall away under him. He fell out of sight.

For a mad instant, while Fradin kicked and hauled me away from that transmitter, the laboratory was hideous with the blast of thunder.

Then another murderous crash came, and....

Then there was silence. Utter silence. The only sound was Fradin fighting for his breath. I looked across the room. The transmitter was gone. It just wasn't there any more. Under it, in the floor of the room was a neat, round hole. All the mass of wires that had led into it were neatly severed. Wires came from the transformer to where the hole began, then stopped.

Marvak wasn't there. Marvak was gone.

Suddenly I turned to Fradin. "You—" I gulped. "You were afraid this would happen. My God, man, what was it?"

"It was," he answered, "a hole in the fourth dimension."

Then I got it. He had been trying to tell that convention of radio engineers that radio waves were transmitted through the fourth dimension, not through the "ether." He had been able to prove his point but he had refused because he knew that this would happen.

"But even if radio waves do pass through the fourth dimension, nothing like this has ever happened," I stammered.

"Ordinary broadcasting stations do not put enough power through their transmitters to open this hole," he explained. "It takes power to do it, lots of power. I had calculated how much power it would take. There was a red mark on the input meter of the transmitter. That red line marked the critical point. If more power was put through the transmitter, it would break down the fabric of space between this dimension and the fourth dimension. I knew it would happen. That's why I refused to make a demonstration for the benefit of my skeptical compatriots. If I told them what I had discovered, proved I had discovered it, some fool would be sure to try it, with disastrous results."

"But that cold wind," I protested.

"This particular region opens out into what must be interplanetary space in the fourth dimension. That cold wind was simply the cold of outer space rushing through what was in effect a window."

So that was it. There was a hole in space. And space is cold.

"Marvak!" I said weakly.

"Don't mention him," Fradin shuddered. "He was catapulted into the fourth dimension. He's frozen solid by now."

I guess the human race will never have power by radio. Probably we will be able to get along without it. Atomic power seems to be coming along, and it's safe.

I took Fradin to the hospital. That slug through his shoulder had cost him a lot of blood, but he recovered all right, only to discover that the Institute of Radio Engineers had booted him right out of their organization, for making the preposterous claim that radio waves are transmitted through the fourth dimension instead of through the ether. However, he never cared two whoops in hell about that. He knew what he knew. And he was content with that.

*** END OF THE PROJECT GUTENBERG EBOOK THE IMPOSSIBLE
INVENTION ***

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